

MT 206: Extraction and Materials Processing

Course Code	MT 206
Course Name	Extraction and Materials Processing
L-T-P-C	3-2-0-4
Prerequisites	None
Intended for	B.Tech. (MSE)
Mutual Exclusion	EN509: Functional Materials for Energy Engineering
Approval	57th BoA (Oct 2024)

Course Contents

- **Introduction to extraction and materials processing:** Define extraction and material processing, processing methods for different types of materials (Metals, Ceramics, Glass, Polymers, and Composites), the significance of these processes in various industries, including urban mining (waste material management), the importance of sustainable sourcing and responsible extraction practices. Discuss standard processing techniques such as crushing, grinding, smelting, refining, and sorting. Environmental and social impacts of extraction methods. [5 hours]
- **Heat treatment:** Annealing, normalizing, hardenability, hardening, tempering, mechanism of heat removal during quenching, quenching media, residual stresses and quench cracks, martempering and austempering, alloying, effect of alloying on Fe-Fe₃C phase diagram, sintering, and spark plasma sintering. [5 hours]
- **Electrolysis:** Electroplating, electrorefining, extraction, purification, and standard electrode potential (SEP) of metals and metal ions. [2 hours]
- **Grinding/Polishing:** Abrasion, classification of abrasive, principles, factors determining grinding/polishing, grinding (ball milling, mechanical alloying, mechanochemical processing). Wet and dry grinding, chemical comminution-leaching, dissolution, screening, crushing and sizing of comminuted particles, concentration techniques: gravity concentration, magnetic and electrostatic separation, froth floatation factors affecting rate of screening. [5 hours]
- **Thin film deposition:** Definition and significance of thin films, thin film growth techniques (CVD, thermal evaporation, sputtering, and solution process), applications of thin films (semiconductor device fabrication, optoelectronic devices, e.g., solar cells, LEDs, OLEDs, and coatings for corrosion protection & wear resistance.), and lithography. [8 hours]
- **Laser Processing:** Introduction to laser processing, basics of laser technology, significance and applications in industry, direct patterning, laser cutting, laser welding, emerging trends in laser processing (e.g., Ultrafast laser processing, laser-assisted material synthesis laser micropatterning and nanostructuring). [5 hours]

Laboratory Modules

1. Heat treatment: To study the material at extremely high temperatures using the muffle furnace and tube furnace. [3 hours]
2. To study the advanced processing methods (Spark plasma sintering and glass processing), [3 hours]
3. To study the fine finishes or light cuts on metals and other materials (grinding). [3 hours]
4. To study the extraction and purification of metals from raw materials. [3 hours]
5. To study electroplating, electro-refining, and the standard electrode potential (SEP) of metals/metal ions. [3 hours]
6. To study the process of creating a smooth and shiny surface by rubbing it or applying a chemical treatment (Polishing). [3 hours]
7. Thin film deposition:
 - (a) Thin film coating using a solution process (e.g., spin coater) [3 hours]
 - (b) Thin film coating using a vacuum process (e.g., sputtering) [3 hours]
8. Laser lithography: To study the modification of the shape or appearance of a material. Micro-scale pattern development. [3 hours]

Text and Reference Books

1. T.V. Rajan, C.P. Sharma, Ashok Rajan, **Heat Treatment: Principles and Techniques**, PHI Learning.
2. William D. Callister, Jr., David G. Rethwisch, **Fundamentals of Materials Science and Engineering**, John Wiley & Sons
3. Colin Bodsworth & Brian Ralph, **The Extraction and Refining of Metals (Materials Science & Technology)**.
4. William F. Smith, **Materials Science and Engineering**, Tata McGraw-Hill Education.
5. Free M., Moats M., Houlachi G., Asselin E., Allanore A., Yurko J., Wang S, **Electrometallurgy**, John Wiley & Sons, Inc., 2012.

References

1. NA.